

Jinhua CUI

Ph.D. student

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Address: College of Computer, National University of Defense
Technology, Changsha 410073, China

EMPLOYMENT

✓ **January 2021 – present**

School of Computing, National University of Singapore

Research Assistant in Crystal Centre

Advisor: Prateek SAXENA

✓ **June 2019 – December 2020**

School of Computing, National University of Singapore

Research Internship in Crystal Centre

Advisor: Prateek SAXENA

✓ **March 2016 – August 2018**

School of Information Systems, Singapore Management University, Singapore

Research Engineer/Senior RE in Secure Mobile Centre

Advisor: Xuhua DING

EDUCATION

✓ **September 2018 – present**

National University of Defense Technology, Changsha, China

Ph.D. in Computer Science

Advisor: Zhiping CAI

✓ **September 2013 – February 2016**

Xidian University, Xi'an, China (Postgraduate Recommendation)

M.S.E. in Circuits and Systems

Advisor: Jianwei HU

✓ **September 2009 – July 2013**

Gansu Agricultural University, Lanzhou, China

B.S.E. in Electronic & Information Engineering

Advisor: Liquan LIU

RESEARCH INTERESTS

- ✓ System Security, Trusted Computing, Mobile Security and Distributed Storage System

PUBLICATIONS

- Shweta Shinde, *Jinhua Cui**, Satyaki Sen, Pinghai Yuan and Prateek Saxena. “Dynamic

Binary Translation for SGX Enclaves”. arXiv preprint arXiv:2009.01144 (version 1, 2020)

- **Jinhua Cui**, Yuanyuan Zhang, Zhiping Cai, Anfeng Liu and Yangyang Li. “Securing Display Path for Security-Sensitive Applications on Mobile Devices”. Computers, Materials & Continua 55, no. 1 (2018): 17-35.
- Zhangkai Zhang, Zhoujun Li, Chunhe Xia, **Jinhua Cui** and Jinxin Ma. “H-Securebox: A Hardened Memory Data Protection Framework on ARM Devices”. 2018 IEEE International Conference on Data Science in Cyberspace (DSC)
- Zhangkai Zhang, Xuhua Ding, Gene Tsudik, **Jinhua Cui** and Zhoujun Li. “Presence Attestation: The Missing Link in Dynamic Trust Bootstrapping”. ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 89-102. ACM, 2017.

PROJECTS

- **Project0: SiGeX: Exception-handling-based Code Reuse Attacks against SGX Enclaves and the Proposed Model (ongoing work)**
- **Project1: RATEL: Dynamic Binary Translation for SGX Enclaves (ongoing work)**
GitHub: <https://github.com/ratel-enclave/ratel>
- **Project2: A Deletion-Resilient P2P Storage System Based on IPFS (November 2018 – June 2019)**
- **Project3: Fortifying Mobile Platforms with a User-Centric Trust Anchor (<https://smc.smu.edu.sg/>) (March 2016 – August 2018)**
Demo videos: <https://youtu.be/jnjpY3XyfP0> (Presence Attestation)
- **Project4: Testing Security of Applications on Smartphones (September 2013 – January 2016)**

ACTIVITIES AND SERVICES

- ✓ As one of the organizers in XDCTF, assisting faculties to host and make out competitive topics in 2014 and 2015, respectively
Website: <http://ctf.xidian.edu.cn/>
- ✓ External Reviewer
IEEE Transactions on Mobile Computing (2020)
IEEE/ACM Transactions on Networking (2019)
IEEE Transactions on Services Computing (2018)
IEEE Internet of Things of Journal (2018)

COMPUTER SKILLS

- ✓ Intel Software Guard Extension (SGX) Protection Mechanism
- ✓ DynamoRIO, Dynamic Instrumentation Tool Platform (for Runtime Introspection & Optimization, etc.)
- ✓ ARM Virtualization, such as KVM/ARM, Xen Hypervisor
- ✓ Have an empirical understanding of underlying system display on Linux and Android
- ✓ Debugging Linux OS, such as Dstream, ARM DS-5, FastModel, etc.
- ✓ Understanding of popular SoC based on ARM, such as Odroid XU4, Raspberry Pi 2/3
- ✓ Have a good knowledge of ARM TrustZone
- ✓ Have an empirical understanding of Linux OS and low-level hardware, such as CPU, cache, MMU, TLB, etc.
- ✓ Have an elementary knowledge of virtualization technology and SGX on x86
- ✓ Linux Kernel, Hypervisor and TrustZone Programming
- ✓ C/Qt/Python/C++/Android Programming

HONORS AND AWARDS

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|---|--------------------------|-------------------------------|------------------------------|
| ✓ | September 2018 – present | PhD Fellowship | |
| ✓ | November 2015 | Xidian University | The First Prize Scholarship |
| ✓ | November 2014 | Xidian University | Outstanding master Awards |
| ✓ | November 2013 | Xidian University | The First Prize Scholarship |
| ✓ | November 2012 | Gansu Agricultural University | National Endeavor Fellowship |
| ✓ | October 2011 | Gansu Agricultural University | National Scholarship |
| ✓ | October 2010 | Gansu Agricultural University | National Endeavor Fellowship |

REFERENCES

- | | | | |
|---|---|---------------------|---|
| ✓ | Prateek Saxena | Associate Professor | School of Computing, National University of Singapore |
| | Home: https://www.comp.nus.edu.sg/~prateeks/ | | |
| | Email: prateeks@comp.nus.edu.sg | | |
| ✓ | Zhiping CAI | Professor | College of Computer, National University of Defense |
| | Home: http://individual.utoronto.ca/zcai/ | | |
| | Email: zpcai@nudt.edu.cn | | |

- ✓ Shweta Shinde Assistant Professor Department of Computer Science, ETH Zurich
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- ✓ Xuhua DING Associate Professor School of Information Systems, Singapore
Management University
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Email: xhding@smu.edu.sg